

Sam Tang

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Education

University of Waterloo

Bachelor of Computer Science | Cumulative Average: 89.6%

September 2022 – Expected April 2027

- University of Waterloo President's Scholarship of Distinction · Faculty of Mathematics Scholarship

Technical Skills

Languages: C++, C#, Python, JavaScript/TypeScript, Java, SQL, C, HTML/CSS

Frameworks: FastAPI, React.js, .NET, Entity Framework Core, ArkTS, Angular

Tools & Technologies: Git, MS SQL Server, Azure, gRPC, Protobuf, MySQL, Pytest, MLIR, GTest, Redis, Bash

Experience

Cerebras

AI Software Developer Intern

January 2026 – April 2026

- Diagnosed and fixed a long-standing wafer stall regression by designing targeted tests and systematically isolating the faulty subsystem, preventing recurring stalls in model workloads
- Built a weight-corruption fault-injection harness using **MLIR**-based weight buffer analysis, **gRPC/Protobuf** backend integration, and bit-level corruption logic to simulate production-like system failures
- Integrated a new model into the simulated system test environment for the latest runtime architecture, reducing test turnaround time by approximately **80%** by avoiding shared-system queue delays

Huawei

Research Software Developer Intern · Virtual Keyboard Core Development

February 2025 – August 2025

- Delivered MateBook Fold virtual keyboard system features at scale (**200K+** users) by owning development end-to-end from design docs through **C++** implementation, experimentation and release
- Broadened multi-finger touchpad support (**2 new gestures, 9 transitions**) by refactoring the touchpad state machine
- Reduced false taps from **25%** to **5%** and improved gesture success by **27%** through signal filtering and input debouncing
- Enabled on-device automated QA input testing by adding a new input-injection hook into the event-handling pipeline

Research Software Developer Intern · AI/HCI Prototype Development

January 2025 – February 2025

- Prototyped a novel **AI/HCI** interaction system with a **JavaScript Chrome extension** and **ArkTS HarmonyOS** companion app, securing approval for continued product development
- Improved gesture recognition accuracy from **40%** to **80%** by optimizing Dollar Point-Cloud (\$P) matching and storage

AutoTrader.ca

Full-stack Developer Intern

May 2024 – August 2024

- Shipped a new vehicle pre-qualification flow under tight release cycles by implementing **Angular** and **.NET** features across frontend and backend services
- Strengthened **API** security boundaries by implementing **server-to-server authentication** from **API controllers** to **Azure APIM** with **Azure B2C JWT** generation
- Cleaned up production data (**7,000+** invalid users removed) by using **Entity Framework** and **Azure AI Search** to accurately identify and delete invalid accounts

Zomp

Full-stack Developer Intern

April 2023 – August 2023

- Improved backend scalability by migrating the **.NET** backend to **Azure Cloud**, removing storage bottlenecks
- Cut **API** response time by **30%** by designing **REST APIs** that stream data efficiently to and from the **Azure** backend
- Reduced an internal workflow from minutes to seconds by building an in-app tool that converts Excel data into an interactive UI component